PHY 421 Assignment 3

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1 Homework 5 - Problem 1

(1) $f(z) = \tan z$

Singularities: $\tan z = \dfrac{\sin z}{\cos z}$ is singular where $\cos z = 0$, i.e.

$$z_k = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

Nature: At each z_k the zero of $\cos z$ is simple, so $\tan z$ has a simple pole at each z_k.

Residue: For a simple pole z_0,

$$\operatorname{Res}(\tan z, z_0) = \lim_{z\to z_0} (z - z_0)\frac{\sin z}{\cos z}.$$

Since $\cos'(z) = -\sin z$, near z_0 we have $\cos z \approx \cos'(z_0)(z - z_0) = -\sin z_0\,(z - z_0)$, hence

$$\operatorname{Res}(\tan z, z_0) = \frac{\sin z_0}{\cos'(z_0)} = \frac{\sin z_0}{-\sin z_0} = -1.$$

So $\boxed{\operatorname{Res}(\tan z, z_k) = -1}$ for every integer k.

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(3) $f(z) = \dfrac{\cot z}{z}$

Singularities: $\cot z$ is singular at $z = k\pi$ and the factor $1/z$ is singular at $z = 0$. So the singular set is $\{z = 0\} \cup \{k\pi : k \in \mathbb{Z} \setminus \{0\}\}$ (note 0 is already of interest).

Nature at $z = 0$: Use the standard Laurent expansion

$$\cot z = \frac{1}{z} - \frac{z}{3} - \frac{z^3}{45} - \cdots .$$

Thus

$$\frac{\cot z}{z} = \frac{1}{z^2} - \frac{1}{3} - \frac{z^2}{45} - \cdots .$$

So at $z = 0$ there is a pole of order 2 (a double pole). The residue is the coefficient of $1/(z-0)$, which here is 0. Hence

$$\boxed{z = 0 \text{ is a double pole with } \operatorname{Res}\left(\tfrac{\cot z}{z}, 0\right) = 0.}$$

Nature at $z = k\pi \neq 0$: For $z_0 = k\pi \neq 0$, $\cot z$ has a simple pole with residue 1 (from part (2)). Since $1/z$ is analytic and nonzero at z_0, f has a simple pole at z_0 with

$$\operatorname{Res}\left(\frac{\cot z}{z}, z_0\right) = \frac{\operatorname{Res}(\cot z, z_0)}{z_0} = \frac{1}{z_0} = \frac{1}{k\pi}.$$

So $\boxed{\operatorname{Res}\left(\frac{\cot z}{z}, k\pi\right) = \dfrac{1}{k\pi}}$ for $k \neq 0$.

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(5) $\quad f(z) = \dfrac{\csc z}{z^2}$

Singularities: $z = k\pi$ for all integers k.

At $z = 0$: Use the series for $\csc z$:

$$\csc z = \frac{1}{z} + \frac{z}{6} + \frac{7z^3}{360} + \cdots.$$

Hence

$$\frac{\csc z}{z^2} = \frac{1}{z^3} + \frac{1}{6z} + \frac{7z}{360} + \cdots.$$

So $z = 0$ is a pole of order 3 (a triple pole). The residue is the coefficient of $1/z$, namely $\frac{1}{6}$. Thus

$$\boxed{z = 0 \text{ is a pole of order 3 with } \operatorname{Res}\left(\tfrac{\csc z}{z^2}, 0\right) = \tfrac{1}{6}.}$$

At $z_0 = k\pi \neq 0$: $\csc z$ has a simple pole at z_0 with residue $(-1)^k$, and $1/z^2$ is analytic at z_0. So f has a simple pole at z_0 with residue

$$\operatorname{Res}\left(\frac{\csc z}{z^2}, z_0\right) = \frac{(-1)^k}{z_0^2} = \frac{(-1)^k}{(k\pi)^2}.$$

So $\boxed{\operatorname{Res}\left(\frac{\csc z}{z^2}, k\pi\right) = \dfrac{(-1)^k}{(k\pi)^2}}$ for $k \neq 0$.

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(7) $\quad f(z) = \dfrac{\tanh z}{z^2}$

(Recall $\tanh z = \dfrac{\sinh z}{\cosh z}$.)

Singularities: $\tanh z$ is meromorphic with poles where $\cosh z = 0$, i.e.

$$z_k = i\frac{\pi}{2} + ik\pi, \qquad k \in \mathbb{Z}.$$

Also $z = 0$ is special because of the z^{-2} factor.

At $z = 0$: Expand $\tanh z = z - \dfrac{z^3}{3} + \dfrac{2z^5}{15} + \cdots$, so

$$\frac{\tanh z}{z^2} = \frac{1}{z} - \frac{z}{3} + \frac{2z^3}{15} + \cdots.$$

Hence $z = 0$ is a simple pole with residue equal to the coefficient of $1/z$, namely 1. Thus

$$\boxed{z = 0 \text{ is a simple pole with } \operatorname{Res}\left(\tfrac{\tanh z}{z^2}, 0\right) = 1.}$$

At z_0 with $\cosh z_0 = 0$: $\cosh z$ has simple zeros at these z_0, so $\tanh z$ has simple poles there. Let z_0 be such a point. Then $\tanh z$ near z_0 behaves like $\dfrac{\sinh z_0}{\cosh'(z_0)(z - z_0)}$ and since $\cosh'(z) = \sinh z$, we get

$$\operatorname{Res}(\tanh z, z_0) = \frac{\sinh z_0}{\sinh z_0} = 1.$$

Hence $f(z) = \tanh z/z^2$ has a simple pole at z_0 with residue

$$\operatorname{Res}\left(\frac{\tanh z}{z^2}, z_0\right) = \frac{1}{z_0^2}.$$

So $\boxed{\operatorname{Res}\left(\tfrac{\tanh z}{z^2},\, z_0\right) = \dfrac{1}{z_0^2}}$ for every z_0 with $\cosh z_0 = 0$.

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(9) $\quad f(z) = \dfrac{e^{1/z}}{1 - z}$

Singularities: There is a simple pole at $z = 1$ (from denominator). Also $z = 0$ is an essential singularity because of the $e^{1/z}$ factor.

At $z = 1$: This is a simple pole. Residue:

$$\operatorname{Res}\left(\frac{e^{1/z}}{1 - z}, 1\right) = \lim_{z \to 1} (z - 1) \frac{e^{1/z}}{1 - z} = -e^{1/1} = -e.$$

So $\boxed{\operatorname{Res}(f, 1) = -e.}$

At $z = 0$: $z = 0$ is essential. If desired the residue at the essential singularity $z = 0$ can be obtained by expanding into a Laurent series about 0. Write

$$\frac{e^{1/z}}{1 - z} = e^{1/z} \cdot \frac{1}{1 - z} = e^{1/z} \sum_{m=0}^{\infty} z^m = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{1}{n!} z^{m-n}.$$

The coefficient of z^{-1} occurs when $m - n = -1$, i.e. $m = n - 1$ with $n \geq 1$. Thus the residue is

$$\operatorname{Res}(f, 0) = \sum_{n=1}^{\infty} \frac{1}{n!} = e - 1.$$

So $\boxed{z = 0 \text{ is essential with } \operatorname{Res}(f, 0) = e - 1.}$

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(11) $\quad f(z) = \dfrac{1}{e^{pz}-1}$ **(real constant p)**

Singularities: Solutions of $e^{pz} = 1$, i.e.

$$pz = 2\pi i k \quad \Longrightarrow \quad z_k = \frac{2\pi i k}{p}, \qquad k \in \mathbb{Z}.$$

(If $p = 0$ the function degenerates; we assume $p \neq 0$.)

Nature: For each k, $e^{pz}-1$ has a simple zero at z_k (derivative $pe^{pz_k} = p \neq 0$), so f has simple poles at each z_k.

Residue: For a simple pole z_k,

$$\operatorname{Res}\left(\frac{1}{e^{pz}-1}, z_k\right) = \frac{1}{(e^{pz}-1)'|_{z_k}} = \frac{1}{pe^{pz_k}} = \frac{1}{p\cdot 1} = \frac{1}{p}.$$

So $\boxed{\operatorname{Res}\left(\frac{1}{e^{pz}-1}, z_k\right) = \dfrac{1}{p}}$ for every integer k.

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(13) $\quad f(z) = \dfrac{e^{pz}}{\cosh^2 z}$ **(real constant p)**

Singularities: $\cosh z = 0$ precisely at

$$z_k = i\frac{\pi}{2} + ik\pi, \qquad k \in \mathbb{Z}.$$

At each such z_k, $\cosh z$ has a simple zero, hence $\cosh^2 z$ has a zero of order 2. Thus f has a pole of order 2 (a double pole) at each z_k.

Residue at a double pole z_0: For a double pole the residue can be computed by

$$\operatorname{Res}(f, z_0) = \lim_{z\to z_0} \frac{d}{dz}\Big[(z-z_0)^2 f(z)\Big].$$

Write $g(z) = \cosh z$, so near z_0 we have $g(z_0) = 0$ and $g'(z_0) = \sinh z_0 \neq 0$. Then

$$\cosh^2 z = g(z)^2 \approx g'(z_0)^2 (z-z_0)^2,$$

and therefore

$$(z-z_0)^2 \frac{e^{pz}}{\cosh^2 z} \approx \frac{e^{pz}}{g'(z_0)^2}.$$

Differentiating and evaluating at z_0 gives

$$\operatorname{Res}\left(\frac{e^{pz}}{\cosh^2 z}, z_0\right) = \frac{d}{dz}\left(\frac{e^{pz}}{g'(z_0)^2}\right)\Bigg|_{z=z_0} = \frac{p\, e^{pz_0}}{g'(z_0)^2} = \frac{p\, e^{pz_0}}{\sinh^2 z_0}.$$

Since z_0 satisfies $\cosh z_0 = 0$, one can write $z_0 = i\frac{\pi}{2} + ik\pi$, and then $\sinh z_0 \neq 0$; the residue formula above is final and explicit. Thus

$$\boxed{z_0 = i\frac{\pi}{2} + ik\pi \text{ is a double pole with } \operatorname{Res}\left(\frac{e^{pz}}{\cosh^2 z}, z_0\right) = \frac{p\, e^{pz_0}}{\sinh^2 z_0}.}$$

(If desired this can be expressed using exponential/trigonometric forms for $\sinh z_0$ and e^{pz_0}.)

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2 Homework 5 - Problem 2

(ii) $J_\pm = \displaystyle\int_0^{2\pi} \frac{d\theta}{a \pm b\sin\theta}$**,** $\quad a > b \geq 0$**.**

Reduction to part (i). Use the shift $\phi = \theta - \frac{\pi}{2}$. Then $\sin\theta = \cos\phi$ and $d\phi = d\theta$. The integration over one full period is invariant under such shift, so

$$\int_0^{2\pi} \frac{d\theta}{a \pm b\sin\theta} = \int_0^{2\pi} \frac{d\phi}{a \pm b\cos\phi}.$$

Therefore by the result of part (i),

$$\boxed{\int_0^{2\pi} \frac{d\theta}{a \pm b\sin\theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}, \qquad a > b \geq 0.}$$

(iv) $L = \displaystyle\int_{-\infty}^{\infty} \frac{x\sin(kx)}{x^2 + a^2}\,dx$**,** $\quad a > 0,\ -\infty < k < \infty$**.**

This integral is evaluated by considering the Fourier-type integral

$$I(k) = \int_{-\infty}^{\infty} \frac{xe^{ikx}}{x^2 + a^2}\,dx$$

and then taking the imaginary part: $L = \operatorname{Im} I(k)$ because $\sin(kx) = \operatorname{Im} e^{ikx}$.

Step 1: evaluate $I(k)$ by contour integration. We treat the two cases $k > 0$ and $k < 0$ separately (the $k = 0$ case is trivial: the integrand is odd so the integral is 0).

Case $k > 0$. Close the integration contour by a semicircle in the upper half-plane where e^{ikz} decays (since $e^{ik(x+iy)} = e^{ikx}e^{-ky}$ for $y > 0$). The integrand has simple poles at $z = ia$ and $z = -ia$, only $z = ia$ lies in the upper half-plane. The residue at $z = ia$ is (simple pole)

$$\operatorname{Res}\Big(\frac{ze^{ikz}}{z^2 + a^2}, z = ia\Big) = \lim_{z \to ia} (z - ia)\frac{ze^{ikz}}{(z - ia)(z + ia)} = \frac{ia\,e^{ik(ia)}}{2ia} = \frac{e^{-ka}}{2}.$$

By the residue theorem

$$I(k) = 2\pi i \cdot \frac{e^{-ka}}{2} = \pi i\, e^{-ka} \qquad (k > 0).$$

Hence

$$L = \operatorname{Im} I(k) = \operatorname{Im}\left(\pi i e^{-ka}\right) = \pi e^{-ka} \qquad (k > 0).$$

Case $k < 0$. For $k < 0$ close the contour in the lower half-plane (so that e^{ikz} decays for $y < 0$). The only enclosed pole is $z = -ia$. The residue at $z = -ia$ is

$$\operatorname{Res}\Big(\frac{ze^{ikz}}{z^2 + a^2}, z = -ia\Big) = \frac{(-ia)\,e^{ik(-ia)}}{2(-ia)} = \frac{e^{ak}}{2}.$$

Because the contour is clockwise in the lower half-plane the integral over the real axis equals $-2\pi i$ times this residue, hence

$$I(k) = -2\pi i \cdot \frac{e^{ak}}{2} = -\pi i\, e^{ak} \qquad (k < 0).$$

Therefore

$$L = \operatorname{Im} I(k) = \operatorname{Im}\left(-\pi i e^{ak}\right) = -\pi e^{ak} = \pi \operatorname{sgn}(k)\, e^{-a|k|} \qquad (k < 0),$$

since $e^{ak} = e^{-a|k|}$ and $\operatorname{sgn}(k) = -1$ here.

Step 2: combine the two cases into one formula. Putting both cases together (and noting $L = 0$ at $k = 0$) we obtain the compact form

$$\boxed{\int_{-\infty}^{\infty} \frac{x\sin(kx)}{x^2+a^2}\,dx = \pi\ \mathrm{sgn}(k)\, e^{-a|k|}\,, \qquad a>0,\ k\in\mathbb{R},}$$

where $\mathrm{sgn}(k) = \begin{cases} 1, & k>0, \\ 0, & k=0, \\ -1, & k<0. \end{cases}$

(One may equivalently write the result piecewise:

$$\int_{-\infty}^{\infty} \frac{x\sin(kx)}{x^2+a^2}\,dx = \begin{cases} \pi e^{-ak}, & k>0, \\ 0, & k=0, \\ -\pi e^{ak}, & k<0, \end{cases}$$

which is the same as $\pi\,\mathrm{sgn}(k)e^{-a|k|}$.)

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3 Homework 5 - Problem 6

(a) Complex integral representation of the unit step function

We want to show

$$\theta(k) = \lim_{\varepsilon\to 0^+} \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{e^{ikx}}{x-i\varepsilon}\,dx,$$

and evaluate the right-hand side (RHS) for $k>0$, $k<0$ and $k=0$.

Interpretation as a contour integral. Treat the integration variable x as the real part of a complex variable z. For fixed $\varepsilon>0$ the integrand

$$f(z) = \frac{e^{ikz}}{z-i\varepsilon}$$

is meromorphic with a single simple pole at $z = i\varepsilon$ (which lies in the upper half-plane since $\varepsilon>0$). We integrate along the real axis from $-R$ to R and close the contour by a semicircle of radius R either in the upper half-plane or in the lower half-plane depending on the sign of k, then let $R\to\infty$.

Case $k>0$. For $k>0$ and $z = x+iy$ with $y>0$,

$$e^{ikz} = e^{ikx}e^{-ky},$$

so $|e^{ikz}| \le e^{-ky}$ decays exponentially as $y\to+\infty$. Thus we close the contour by a large semicircle in the *upper* half-plane. By Jordan's lemma the contribution of the semicircle tends to 0 as $R\to\infty$. The pole at $z=i\varepsilon$ lies inside the closed contour, so by the residue theorem

$$\int_{-\infty}^{\infty} \frac{e^{ikx}}{x-i\varepsilon}\,dx = \ 2\pi i\ \mathrm{Res}_{z=i\varepsilon}\,\frac{e^{ikz}}{z-i\varepsilon}.$$

Compute the residue (simple pole):

$$\operatorname{Res}_{z=i\varepsilon} \frac{e^{ikz}}{z-i\varepsilon} = \lim_{z\to i\varepsilon} (z-i\varepsilon)\frac{e^{ikz}}{z-i\varepsilon} = e^{ik(i\varepsilon)} = e^{-k\varepsilon}.$$

Hence

$$\int_{-\infty}^{\infty} \frac{e^{ikx}}{x-i\varepsilon}\, dx = 2\pi i\, e^{-k\varepsilon}.$$

Multiply by $\dfrac{1}{2\pi i}$ and take $\varepsilon \to 0^+$ to obtain

$$\lim_{\varepsilon\to 0^+} \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{ikx}}{x-i\varepsilon}\, dx = \lim_{\varepsilon\to 0^+} e^{-k\varepsilon} = 1 \qquad (k>0).$$

Case $k<0$. For $k<0$ and $z=x+iy$ with $y<0$,

$$e^{ikz} = e^{ikx}e^{-ky},$$

and now for $y<0$ (i.e. in the lower half-plane) $-ky$ is positive so e^{ikz} decays as $|y|\to\infty$ in the lower half-plane. Thus we close the contour in the *lower* half-plane. The pole at $z=i\varepsilon$ is in the upper half-plane and therefore *not* enclosed; by Cauchy's theorem the integral over the closed contour is zero, and the semicircular contribution vanishes by Jordan's lemma. Hence

$$\int_{-\infty}^{\infty} \frac{e^{ikx}}{x-i\varepsilon}\, dx = 0 \qquad (k<0),$$

and therefore

$$\lim_{\varepsilon\to 0^+} \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{ikx}}{x-i\varepsilon}\, dx = 0 \qquad (k<0).$$

Case $k=0$. When $k=0$ the integral reduces to

$$\frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{1}{x-i\varepsilon}\, dx.$$

Symmetry and the usual distributional interpretation (or taking the average of the limits from $k\to 0^{\pm}$) give the value $\frac{1}{2}$. More rigorously one can evaluate the principal value

$$\text{p. v.} \int_{-\infty}^{\infty} \frac{dx}{x-i0} = i\pi,$$

so that $\dfrac{1}{2\pi i}(i\pi) = \frac{1}{2}$. Thus the limiting value at $k=0$ is $\frac{1}{2}$.

Putting the three cases together we recover the Heaviside (unit step) function:

$$\theta(k) = \begin{cases} 1, & k>0, \\ \dfrac{1}{2}, & k=0, \\ 0, & k<0, \end{cases}$$

and hence the claimed representation is proved:

$$\boxed{\theta(k) = \lim_{\varepsilon\to 0^+} \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{ikx}}{x-i\varepsilon}\, dx}$$

(b) Complex integral representation of the Kronecker delta

We need to show

$$\delta_{m,n} = \frac{1}{2\pi i} \oint_C z^{\,m-n-1}\, dz,$$

where $m, n \in \mathbb{Z}$ and C is any simple closed contour (circle) centered at the origin.

The integrand is z^{m-n-1}. Consider the exponent

$$\ell := m - n - 1.$$

- If $\ell = -1$, i.e. $m - n = 0$ or $m = n$, then $z^{m-n-1} = z^{-1}$ and

  $$\oint_C z^{-1}\, dz = 2\pi i,$$

  since the residue of z^{-1} at $z = 0$ is 1.

- If $\ell \neq -1$, i.e. $m \neq n$, then z^{ℓ} is analytic everywhere inside and on C (in particular at $z = 0$) and Cauchy's theorem implies

  $$\oint_C z^{\ell}\, dz = 0.$$

Combining these two cases,

$$\frac{1}{2\pi i} \oint_C z^{m-n-1}\, dz = \begin{cases} 1, & m = n, \\ 0, & m \neq n, \end{cases}$$

which is exactly $\delta_{m,n}$. Hence

$$\boxed{\delta_{m,n} = \frac{1}{2\pi i} \oint_C z^{\,m-n-1}\, dz\ .}$$

4 Homework 5 - Problem 8

(c) Evaluate $\int_0^\infty e^{-i(\omega \pm \omega_0)t}\, dt$ as a distribution

Let

$$k := \omega \pm \omega_0,$$

so the integral becomes $I(k) = \int_0^\infty e^{-ikt}\, dt$. The oscillatory integral is understood in the sense of distributions and is obtained by the standard damping regularization:

$$I(k) = \lim_{\varepsilon \downarrow 0} \int_0^\infty e^{-ikt} e^{-\varepsilon t}\, dt = \lim_{\varepsilon \downarrow 0} \int_0^\infty e^{-(\varepsilon + ik)t}\, dt.$$

For each fixed $\varepsilon > 0$ the integral converges absolutely and equals

$$\int_0^\infty e^{-(\varepsilon + ik)t}\, dt = \frac{1}{\varepsilon + ik}.$$

Thus, as a distribution,

$$I(k) = \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon + ik}.$$

We now express $\dfrac{1}{\varepsilon + ik}$ in a form to which the Sokhotski–Plemelj identity applies. Note

$$\frac{1}{\varepsilon + ik} = \frac{1}{i} \cdot \frac{1}{k - i\varepsilon} = -i \cdot \frac{1}{k - i\varepsilon}.$$

Applying the distributional limit (with $x = k$ and $-i\varepsilon$ corresponding to the "$-i0$" prescription),

$$\lim_{\varepsilon \downarrow 0} \frac{1}{k - i\varepsilon} = \mathrm{P.\,V.}\, \frac{1}{k} + i\pi\delta(k).$$

Therefore

$$I(k) = \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon + ik} = -i\left(\mathrm{P.\,V.}\, \frac{1}{k} + i\pi\delta(k) \right) = \pi\delta(k) - i\ \mathrm{P.\,V.}\, \frac{1}{k}.$$

Returning to the original variable $k = \omega \pm \omega_0$, we obtain the distributional identity

$$\boxed{\int_0^\infty e^{-i(\omega \pm \omega_0)t}\, dt = \pi\, \delta(\omega \pm \omega_0) - i\ \mathrm{P.\,V.}\, \frac{1}{\omega \pm \omega_0}}$$

(variant forms differing by sign in the principal-value term can appear depending on the sign convention used in the Sokhotski–Plemelj identity; the displayed form above is the direct consequence of the regularization $e^{-\varepsilon t}$.)

Remark on the sign: the alternative (equivalent) representation sometimes written in textbooks is

$$\int_0^\infty e^{-i(\omega \pm \omega_0)t}\, dt = i\ \mathrm{P.\,V.}\, \frac{1}{-(\omega \pm \omega_0)} + \pi\delta(\omega \pm \omega_0),$$

and since $\mathrm{P.\,V.}\, \frac{1}{-x} = -\,\mathrm{P.\,V.}\, \frac{1}{x}$ this is algebraically equivalent to the boxed formula above. One must be careful with consistent sign conventions when comparing different references.

(d) Representation of the delta distribution

We want to show

$$\delta(x - x_0) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \left[\frac{1}{x - x_0 - i\varepsilon} - \frac{1}{x - x_0 + i\varepsilon} \right].$$

Start from the Sokhotski–Plemelj identity applied with $y = x - x_0$:

$$\lim_{\varepsilon \downarrow 0} \frac{1}{y \pm i\varepsilon} = \mathrm{P.\,V.}\, \frac{1}{y} \mp i\pi\delta(y).$$

Write the two limits explicitly:

$$\lim_{\varepsilon \downarrow 0} \frac{1}{y - i\varepsilon} = \mathrm{P.\,V.}\, \frac{1}{y} + i\pi\delta(y), \qquad \lim_{\varepsilon \downarrow 0} \frac{1}{y + i\varepsilon} = \mathrm{P.\,V.}\, \frac{1}{y} - i\pi\delta(y).$$

Subtract the second from the first:

$$\lim_{\varepsilon \downarrow 0} \left(\frac{1}{y - i\varepsilon} - \frac{1}{y + i\varepsilon} \right) = 2i\pi\delta(y).$$

Dividing both sides by $2\pi i$ yields

$$\delta(y) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \left(\frac{1}{y - i\varepsilon} - \frac{1}{y + i\varepsilon} \right).$$

Putting back $y = x - x_0$ gives the claimed formula:

$$\boxed{\delta(x - x_0) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \left[\frac{1}{x - x_0 - i\varepsilon} - \frac{1}{x - x_0 + i\varepsilon} \right].}$$

5 Homework 5 - Problem 9

1. Set notation. Define

$$\mathbf{R} \equiv \mathbf{r} - \mathbf{r}', \qquad R \equiv |\mathbf{R}|.$$

Because the integrand depends on $\mathbf{k}'$ only through k' and the dot product $\mathbf{k}' \cdot \mathbf{R}$, we choose spherical coordinates in k'-space with the polar axis along $\mathbf{R}$. Then $\mathbf{k}' \cdot \mathbf{R} = k'R\cos\theta$ and $d^3k' = k'^2 dk'\, d\Omega$.

2. Perform the angular integral. The angular part is

$$\int_{S^2} e^{i\mathbf{k}'\cdot\mathbf{R}}\, d\Omega = \int_0^{2\pi} d\phi \int_0^{\pi} e^{ik'R\cos\theta} \sin\theta\, d\theta = 4\pi \frac{\sin(k'R)}{k'R}.$$

Hence

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{(2\pi)^3} \int_0^\infty k'^2 dk' \, \frac{4\pi \, \sin(k'R)}{k'R} \, \frac{1}{k^2 - k'^2}.$$

Simplify the prefactors:

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{2\pi^2 R} \int_0^\infty \frac{k' \sin(k'R)}{k^2 - k'^2} \, dk'.$$

3. Introduce the $i\varepsilon$ prescription. The denominator has poles on the real k'-axis at $k' = \pm k$. To define the integral unambiguously we displace the poles slightly off the axis:

$$\frac{1}{k^2 - k'^2} \longrightarrow \frac{1}{k^2 - k'^2 \mp i\varepsilon},$$

where the choice of $\mp i\varepsilon$ determines whether we obtain an outgoing $(+ikR)$ or incoming $(-ikR)$ spherical wave. We will carry the sign through and explain the outcome at the end.

Thus

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{2\pi^2 R} \int_0^\infty \frac{k' \sin(k'R)}{k^2 - k'^2 \mp i\varepsilon} \, dk'.$$

4. Turn the sine into exponentials and extend to $-\infty$–∞. Write $\sin(k'R) = \frac{1}{2i}\big(e^{ik'R} - e^{-ik'R}\big)$. Then

$$\int_0^\infty \frac{k' \sin(k'R)}{k^2 - k'^2 \mp i\varepsilon} \, dk' = \frac{1}{2i} \int_0^\infty k' \frac{e^{ik'R} - e^{-ik'R}}{k^2 - k'^2 \mp i\varepsilon} \, dk'.$$

Combine the two half-line integrals into a single integral over the whole real axis by the substitution $k' \mapsto -k'$ in the second term. One obtains

$$\int_0^\infty \frac{k' \sin(k'R)}{k^2 - k'^2 \mp i\varepsilon} \, dk' = \frac{1}{4i} \int_{-\infty}^\infty \frac{k' \, e^{ik'R}}{k^2 - k'^2 \mp i\varepsilon} \, dk'.$$

Therefore

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi^2 i R} \int_{-\infty}^\infty \frac{k' \, e^{ik'R}}{k^2 - k'^2 \mp i\varepsilon} \, dk'. \qquad (*)$$

5. Evaluate the real–line integral by contour integration. Consider the integral

$$I \;=\; \int_{-\infty}^\infty \frac{k' \, e^{ik'R}}{k^2 - k'^2 \mp i\varepsilon} \, dk'.$$

Assume $R > 0$. For convergence of the exponential at large $|k'|$ we choose to close the contour in the upper half complex k'-plane because $e^{ik'R}$ decays exponentially when $\operatorname{Im} k' > 0$.

The integrand has simple poles where the denominator vanishes:

$$k^2 - k'^2 \mp i\varepsilon = 0 \quad \Longrightarrow \quad k' = \pm k \text{ shifted slightly off the real axis.}$$

With the $\mp i\varepsilon$ prescription the pole positions are:

$k' = +k$ (shifted to $+k+i0$ for the "$-i\varepsilon$" choice), $\qquad k' = -k$ (shifted to $-k-i0$ or $-k+i0$ depending on sign).

For the common physical choice that yields an *outgoing* spherical wave one takes the denominator $k^2 - k'^2 - i\varepsilon$. Then the pole at $k' = +k + i0$ lies in the upper half-plane and the pole at $-k - i0$ lies in the lower half-plane. Thus when we close the contour in the upper half-plane we enclose only the pole at $k' = k + i0$.

Compute the residue at a simple pole $k' = k_p$. For the function $\dfrac{k'e^{ik'R}}{k^2 - k'^2}$,

$$\mathrm{Res}_{k'=k_p} = \lim_{k'\to k_p} (k' - k_p)\frac{k'e^{ik'R}}{k^2 - k'^2} = \frac{k_p e^{ik_pR}}{-2k_p} = -\frac{1}{2}e^{ik_pR}.$$

For the pole $k_p = k$ the residue is $-\frac{1}{2}e^{ikR}$. By the residue theorem (closing in the upper half-plane)

$$I = 2\pi i \cdot \big(\text{sum of residues inside contour}\big) = 2\pi i \left(-\tfrac{1}{2}e^{ikR}\right) = -i\pi e^{ikR}.$$

6. Insert back into $(*)$ and simplify. Using the value of I we get

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi^2 iR}\, I = \frac{1}{4\pi^2 iR}\left(-i\pi e^{ikR}\right) = -\frac{1}{4\pi R}e^{ikR}.$$

7. Interpret the sign and the $i\varepsilon$ choice. The specific choice we made for the denominator,

$$\frac{1}{k^2 - k'^2 - i\varepsilon},$$

produces the *outgoing* spherical wave $-\dfrac{1}{4\pi}\dfrac{e^{ikR}}{R}$. If instead one had chosen the opposite prescription $\dfrac{1}{k^2 - k'^2 + i\varepsilon}$, the pole enclosed by the upper half-plane contour would be different and the result would be the *incoming* spherical wave $-\dfrac{1}{4\pi}\dfrac{e^{-ikR}}{R}$.

8. Final answer. Therefore, with the $i\varepsilon$ prescription,

$$\boxed{G(\mathbf{r}, \mathbf{r}') = \int \frac{d^3k'}{(2\pi)^3}\,\frac{e^{i\mathbf{k}'\cdot(\mathbf{r}-\mathbf{r}')}}{k^2 - (k')^2 \mp i\varepsilon} \;=\; -\frac{1}{4\pi}\,\frac{e^{\pm ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}},$$

where the upper sign in the exponential corresponds to the choice $-i\varepsilon$ (outgoing wave) and the lower sign to $+i\varepsilon$ (incoming wave).

This $G(\mathbf{r}, \mathbf{r}')$ is precisely the free-particle Green's function (propagator) in three dimensions.

6 Homework 6 - Problem 1

The function $f(x)$ is an odd function about $x = 0$:

$$f(-x) = -f(x).$$

Hence its Fourier series on the interval $(-L, L)$ has only sine terms (no constant term and no cosine terms). The standard Fourier series on $(-L, L)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\Big(\frac{n\pi x}{L}\Big) + b_n \sin\Big(\frac{n\pi x}{L}\Big) \right],$$

with coefficients

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos\Big(\frac{n\pi x}{L}\Big)\, dx, \qquad b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin\Big(\frac{n\pi x}{L}\Big)\, dx.$$

Because f is odd, $a_0 = 0$ and $a_n = 0$ for all n. Compute b_n:

$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin\Big(\frac{n\pi x}{L}\Big)\, dx = \frac{2}{L} \int_{0}^{L} f(x) \sin\Big(\frac{n\pi x}{L}\Big)\, dx,$$

where we used oddness to reduce to $(0, L)$. On $(0, L)$ we have $f(x) = C$, so

$$b_n = \frac{2}{L} \int_0^L C \sin\Big(\frac{n\pi x}{L}\Big)\, dx = \frac{2C}{L} \cdot \frac{L}{n\pi} \big(1 - \cos(n\pi)\big) = \frac{2C}{n\pi} \big(1 - (-1)^n\big).$$

Thus

$$b_n = \begin{cases} 0, & n \text{ even}, \\ \dfrac{4C}{n\pi}, & n \text{ odd}. \end{cases}$$

Therefore the Fourier series contains only odd n sine terms. Writing $n = 2m+1$ for $m = 0, 1, 2, \ldots$ we get

$$\boxed{f(x) = \sum_{m=0}^{\infty} \frac{4C}{(2m+1)\pi} \sin\Big(\frac{(2m+1)\pi x}{L}\Big).}$$

Remarks (Gibbs phenomenon)

- The partial sums of the series converge to $f(x)$ at every point of continuity and to the average of left and right limits at discontinuities (here at $x = \ell L$ for integer ℓ).
- Near the jump discontinuities (for example at $x = 0$), the partial sums exhibit oscillatory overshoot/undershoot which does not vanish as more terms are taken; the overshoot approaches the Gibbs constant $\approx 0.08949\ldots$ times the jump height. This is the Gibbs phenomenon.
- For plotting one typically shows partial sums for $n_{\max} = 50, 100, 200$ over several periods; the series (normalized by C) reproduces the square-wave shape with ringing near the jumps.

7 Homework 6 - Problem 2

We expand $f(x)$ in a Fourier sine series on $[0,\pi]$ (extend f as an odd function to $[-\pi,\pi]$). The sine-series coefficients are

$$b_n = \frac{2}{\pi}\int_0^\pi f(x)\sin(nx)\,dx = \frac{2}{\pi}\int_0^\pi x(\pi-x)\sin(nx)\,dx, \qquad n = 1,2,3,\dots$$

Compute the integral (by integration by parts or known integrals). One convenient closed form is

$$\int_0^\pi x(\pi-x)\sin(nx)\,dx = \frac{2\big(1-(-1)^n\big)}{n^3}.$$

(You may verify this by two integrations by parts or symbolic integration.)

Hence

$$b_n = \frac{2}{\pi}\cdot\frac{2\big(1-(-1)^n\big)}{n^3} = \frac{4}{\pi}\,\frac{1-(-1)^n}{n^3}.$$

Thus $b_n = 0$ for even n, and for odd $n = 2m+1$,

$$b_{2m+1} = \frac{8}{\pi(2m+1)^3}, \qquad m = 0,1,2,\dots$$

Therefore the sine series is

$$\boxed{x(\pi-x) = \frac{8}{\pi}\sum_{m=0}^{\infty}\frac{\sin\big((2m+1)x\big)}{(2m+1)^3}, \qquad 0\le x\le\pi.}$$

Remarks on plotting

- The right-hand series converges rapidly (coefficients $\propto 1/(2m+1)^3$), so a few terms already produce an accurate approximation on $[0,\pi]$.
- Plot $f(x)$ and a partial sum of the series (say first 5–20 odd terms) on the same axes to compare.

8 Homework 6 - Problem 3

Solution (a): Fourier series on $(-\pi,\pi)$

We expand $\cos(\omega_0 t)$ on $(-\pi,\pi)$ in a cosine Fourier series (the function is even, so only cosines appear):

$$\cos(\omega_0 t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos(nt),$$

with

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}\cos(\omega_0 t)\,dt, \qquad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\cos(\omega_0 t)\cos(nt)\,dt.$$

Compute a_0:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}\cos(\omega_0 t)\,dt = \frac{2}{\pi}\int_0^\pi\cos(\omega_0 t)\,dt = \frac{2}{\pi}\cdot\frac{\sin(\pi\omega_0)}{\omega_0}.$$

Hence

$$\frac{a_0}{2} = \frac{\sin(\pi\omega_0)}{\pi\omega_0}.$$

Now compute a_n for $n \geq 1$. Use the product-to-sum identity:

$$\cos(\omega_0 t)\cos(nt) = \tfrac{1}{2}\big[\cos((\omega_0+n)t) + \cos((\omega_0-n)t)\big].$$

Integrate from $-\pi$ to π:

$$\int_{-\pi}^{\pi} \cos((\omega_0 \pm n)t)\,dt = \frac{2\sin(\pi(\omega_0 \pm n))}{\omega_0 \pm n} = 2\frac{\sin(\pi\omega_0)\cos(n\pi)}{\omega_0 \pm n},$$

because $\sin(\pi(\omega_0 \pm n)) = \sin(\pi\omega_0)\cos(\pi n) + \cos(\pi\omega_0)\sin(\pi n) = \sin(\pi\omega_0)(-1)^n$. Thus

$$a_n = \frac{1}{\pi}\left[\frac{\sin(\pi\omega_0)(-1)^n}{\omega_0+n} + \frac{\sin(\pi\omega_0)(-1)^n}{\omega_0-n}\right] = \frac{2\omega_0\sin(\pi\omega_0)}{\pi}\cdot\frac{(-1)^n}{\omega_0^2-n^2}.$$

Therefore

$$\cos(\omega_0 t) = \frac{\sin(\pi\omega_0)}{\pi\omega_0} + \sum_{n=1}^{\infty}\frac{2\omega_0\sin(\pi\omega_0)}{\pi}\cdot\frac{(-1)^n\cos(nt)}{\omega_0^2-n^2}.$$

Factor $\dfrac{\sin(\pi\omega_0)}{\pi\omega_0}$:

$$\cos(\omega_0 t) = \frac{\sin(\pi\omega_0)}{\pi\omega_0}\left[1 + 2\omega_0^2\sum_{n=1}^{\infty}\frac{(-1)^n\cos(nt)}{\omega_0^2-n^2}\right],$$

which is the formula to be shown (valid for $\omega_0 \notin \mathbb{Z}$; when ω_0 equals an integer one must treat resonant terms by limiting procedures).

Solution (b): Fourier transform of $\cos(\omega_0 t)$

We use the transform convention

$$F(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\cos(\omega_0 t)\,e^{-i\omega t}\,dt.$$

Write $\cos(\omega_0 t) = \frac{1}{2}\big(e^{i\omega_0 t} + e^{-i\omega_0 t}\big)$. Using the distributional identity $\int_{-\infty}^{\infty} e^{i\alpha t}e^{-i\omega t}\,dt = 2\pi\delta(\omega-\alpha)$ we obtain

$$\begin{aligned} F(\omega) &= \frac{1}{\sqrt{2\pi}}\cdot\frac{1}{2}\Big[\int_{-\infty}^{\infty} e^{i\omega_0 t}e^{-i\omega t}\,dt + \int_{-\infty}^{\infty} e^{-i\omega_0 t}e^{-i\omega t}\,dt\Big] \\ &= \frac{1}{\sqrt{2\pi}}\cdot\frac{1}{2}\Big[2\pi\delta(\omega-\omega_0) + 2\pi\delta(\omega+\omega_0)\Big] = \sqrt{\frac{\pi}{2}}\,\big[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\big]. \end{aligned}$$

Hence

$$\boxed{\mathcal{F}\{\cos(\omega_0 t)\}(\omega) = \sqrt{\frac{\pi}{2}}\,\big[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\big].}$$

Solution (c): Plots / sketch

- The Fourier series derived in part (a) represents $\cos(\omega_0 t)$ on $(-\pi,\pi)$ as a discrete cosine expansion. If ω_0 is *not* an integer the series converges pointwise to $\cos(\omega_0 t)$ on $(-\pi,\pi)$.
- The Fourier transform $F(\omega)$ is a pair of symmetric delta spikes located at $\omega = \pm\omega_0$ (their weights given above). A sketch of $F(\omega)$ is therefore two impulses at $\pm\omega_0$ with equal amplitudes $\sqrt{\pi/2}$.

- If you plot partial sums of the series (finite $n_{\max}$) you will see how the partial-sum approximation approaches $\cos(\omega_0 t)$ on the interval but exhibits the usual truncation artefacts near the boundaries of the period when viewed periodically.

Notes on plotting and numerical checks:

- For Problem 1: to illustrate Gibbs phenomenon plot the partial sums

$$S_N(x) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{N} \frac{4C}{n\pi} \sin\Big(\frac{n\pi x}{L}\Big)$$

  for $N = 50, 100, 200$ (or equivalently $n_{\max}$ number of odd terms) over several periods.

- For Problem 2: the cubic decay $(2m+1)^{-3}$ makes convergence fast; plot the exact $x(\pi - x)$ and the partial sum with first e.g. 10 odd terms.

- For Problem 3: the Fourier transform is a distribution (delta functions); to visualize in a plotting program one typically draws spikes at $\pm\omega_0$ with height proportional to the amplitude $\sqrt{\pi/2}$.

9 Homework 6 - Problem 5

(a) Fourier series of the periodic function f(x)

We derive the Fourier series for e^x on $(-\pi, \pi)$. The derivation for e^{-x} is analogous and the final expression for both signs is compactly written as above.

We use the complex exponential Fourier coefficients

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x e^{-inx}\, dx, \qquad n \in \mathbb{Z},$$

and then convert to real Fourier coefficients. Compute the integral:

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1-in)x}\, dx = \frac{1}{2\pi} \left[\frac{e^{(1-in)x}}{1-in} \right]_{x=-\pi}^{\pi} \\ &= \frac{1}{2\pi} \frac{e^{(1-in)\pi} - e^{-(1-in)\pi}}{1-in} = \frac{1}{2\pi} \frac{e^{\pi} e^{-in\pi} - e^{-\pi} e^{in\pi}}{1-in}. \end{aligned}$$

Factor $e^{-in\pi} = (-1)^n$ and $e^{in\pi} = (-1)^n$ to get

$$c_n = \frac{(-1)^n}{2\pi} \frac{e^{\pi} - e^{-\pi}}{1-in} = \frac{(-1)^n \sinh\pi}{\pi} \cdot \frac{1}{1-in}.$$

Now the real Fourier series on $(-\pi, \pi)$ is

$$e^x = \frac{a_0}{2} + \sum_{n=1}^{\infty} \big(a_n \cos(nx) + b_n \sin(nx)\big),$$

with the relations (from the complex coefficients)

$$a_n = c_n + c_{-n}, \qquad b_n = i(c_n - c_{-n}), \qquad a_0 = 2c_0.$$

Using the formula for c_n we compute

$$c_{-n} = \frac{(-1)^{-n}\sinh\pi}{\pi}\cdot\frac{1}{1+in} = \frac{(-1)^{n}\sinh\pi}{\pi}\cdot\frac{1}{1+in}.$$

Hence

$$a_0 = 2c_0 = 2\cdot\frac{\sinh\pi}{\pi}\cdot\frac{1}{1-i0} = \frac{2\sinh\pi}{\pi},$$

$$a_n = c_n + c_{-n} = \frac{(-1)^n\sinh\pi}{\pi}\Big(\frac{1}{1-in}+\frac{1}{1+in}\Big) = \frac{2(-1)^n\sinh\pi}{\pi}\cdot\frac{1}{1+n^2},$$

$$b_n = i(c_n - c_{-n}) = i\frac{(-1)^n\sinh\pi}{\pi}\Big(\frac{1}{1-in}-\frac{1}{1+in}\Big) = -\frac{2(-1)^n n\sinh\pi}{\pi}\cdot\frac{1}{1+n^2}.$$

Therefore

$$\begin{aligned} e^x &= \frac{a_0}{2} + \sum_{n=1}^{\infty}\Big[\frac{2(-1)^n\sinh\pi}{\pi(1+n^2)}\cos(nx) - \frac{2(-1)^n n\sinh\pi}{\pi(1+n^2)}\sin(nx)\Big] \\ &= \frac{\sinh\pi}{\pi}\left[1 + 2\sum_{n=1}^{\infty}\frac{(-1)^n\big(\cos(nx) - n\sin(nx)\big)}{1+n^2}\right]. \end{aligned}$$

Replacing $x \mapsto -x$ and/or repeating the calculation for e^{-x} yields the companion result; combining both signs compactly:

$$\boxed{e^{\pm x} = \frac{\sinh\pi}{\pi}\left[1 + 2\sum_{n=1}^{\infty}\frac{(-1)^n\big(\cos(nx) \mp n\sin(nx)\big)}{1+n^2}\right].}$$

(b) Fourier series of $\sinh x$ and $\cosh x$

Use the linear combinations

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2},$$

and the formula for $e^{\pm x}$ above. Since in the bracket the cosine parts are the same for e^x and e^{-x} while the sine parts change sign, simple algebra gives:

$$\cosh x = \frac{\sinh\pi}{\pi}\left[1 + 2\sum_{n=1}^{\infty}\frac{(-1)^n\cos(nx)}{1+n^2}\right],$$

$$\sinh x = \frac{1}{2}\big(e^x - e^{-x}\big) = -\frac{\sinh\pi}{\pi}\,2\sum_{n=1}^{\infty}\frac{(-1)^n n\sin(nx)}{1+n^2}.$$

It is common to write the sinh series with a flipped sign inside the alternating factor:

$$\boxed{\cosh x = \frac{\sinh\pi}{\pi}\left[1 + 2\sum_{n=1}^{\infty}\frac{(-1)^n\cos(nx)}{1+n^2}\right], \qquad \sinh x = -\frac{2\sinh\pi}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n n\sin(nx)}{1+n^2}.}$$

(Equivalently $\sinh x = \frac{2\sinh\pi}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1} n\sin(nx)}{1+n^2}$.)

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10 Homework 6 - Problem 6

(a) If $f(x)$ is real, what restriction is imposed on the coefficients c_n?

If $f(x)$ is real for all x then its Fourier series equals its complex conjugate:

$$f(x) = \overline{f(x)} = \sum_{n=-\infty}^{\infty} \overline{c_n}\, e^{-inx} = \sum_{m=-\infty}^{\infty} \overline{c_{-m}}\, e^{imx}.$$

Comparing coefficients of e^{imx} yields

$$\boxed{c_{-m} = \overline{c_m} \quad \text{for all } m.}$$

Equivalently $c_{-n} = \overline{c_n}$.

(b) If $f(-x) = f(x)$ (even function), what restriction is imposed on c_n?

Evenness implies

$$f(-x) = \sum_n c_n e^{-inx} = \sum_n c_n e^{inx} = f(x).$$

Comparing with $f(x) = \sum_n c_n e^{inx}$ we get $c_n = c_{-n}$ for all n. Thus

$$\boxed{c_{-n} = c_n \quad \text{(evenness).}}$$

If in addition f is real then by (a) c_n are real and even in index.

*(c) If $f(-x) = -f(x)$ (odd function), what restriction is imposed on c_n?
Oddness implies

$$f(-x) = -f(x) \quad \Longrightarrow \quad \sum_n c_n e^{-inx} = -\sum_n c_n e^{inx}.$$

Replacing index $n \mapsto -n$ on the left gives $\sum_n c_{-n} e^{inx} = -\sum_n c_n e^{inx}$, hence

$$\boxed{c_{-n} = -c_n \quad \text{(oddness).}}$$

If f is also real then $c_{-n} = -c_n = \overline{c_n}$, so each c_n is purely imaginary and antisymmetric in index.

(d) Parseval's identity's form

Start from

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

Compute the mean square:

$$\frac{1}{2\pi}\int_0^{2\pi} |f(x)|^2\, dx = \frac{1}{2\pi}\int_0^{2\pi} \Big(\sum_n c_n e^{inx}\Big)\Big(\sum_m \overline{c_m} e^{-imx}\Big)\, dx$$

$$= \sum_n \sum_m c_n \overline{c_m}\, \frac{1}{2\pi}\int_0^{2\pi} e^{i(n-m)x}\, dx.$$

The orthogonality integral equals 1 when $n = m$ and 0 otherwise, so only diagonal terms survive:

$$\frac{1}{2\pi}\int_0^{2\pi} |f(x)|^2\,dx = \sum_{n=-\infty}^{\infty} c_n\overline{c_n} = \sum_{n=-\infty}^{\infty} |c_n|^2.$$

Split the double–sided sum:

$$\sum_{n=-\infty}^{\infty} |c_n|^2 = |c_0|^2 + \sum_{n=1}^{\infty}\left(|c_n|^2 + |c_{-n}|^2\right) = |c_0|^2 + 2\sum_{n=1}^{\infty}|c_n|^2,$$

since $|c_{-n}| = |c_n|$. Thus Parseval's identity in the requested form is

$$\boxed{\frac{1}{2\pi}\int_0^{2\pi} [f(x)]^2\,dx = c_0^2 + 2\sum_{n=1}^{\infty}|c_n|^2\,.}$$

Remark. For a real-valued f, one may replace $|c_n|^2$ by the squared real coefficients when using the sine/cosine form; the complex exponential formulation is concise and often most convenient for proofs.

11 Homework 6 - Problem 13

The position of an underdamped oscillator is

$$x(t) = e^{-\gamma t}\cos(\omega_0 t)\,\theta(t),$$

where $\gamma > 0$ is the damping factor, ω_0 the undamped natural frequency, and $\theta(t)$ the unit step function.

(a) Fourier transform

We compute

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-i\omega t}\,dt = \int_0^{\infty} e^{-\gamma t}\cos(\omega_0 t)\,e^{-i\omega t}\,dt,$$

since $\theta(t)$ restricts the integral to $[0, \infty)$.

Write $\cos(\omega_0 t) = \frac{1}{2}\left(e^{i\omega_0 t} + e^{-i\omega_0 t}\right)$. Then

$$\begin{aligned} X(\omega) &= \frac{1}{2}\int_0^{\infty} e^{-\gamma t}\left(e^{i\omega_0 t} + e^{-i\omega_0 t}\right)e^{-i\omega t}\,dt \\ &= \frac{1}{2}\int_0^{\infty}\left(e^{-(\gamma+i(\omega-\omega_0))t} + e^{-(\gamma+i(\omega+\omega_0))t}\right)dt. \end{aligned}$$

Each integral is an exponential integral of the form $\int_0^\infty e^{-at}\,dt = 1/a$ for $\mathrm{Re}(a) > 0$. Thus

$$X(\omega) = \frac{1}{2}\left[\frac{1}{\gamma + i(\omega-\omega_0)} + \frac{1}{\gamma + i(\omega+\omega_0)}\right].$$

We can algebraically manipulate this expression into the form given in the problem statement. Combine the two fractions (symbolically) and simplify:

$$X(\omega) = \frac{\gamma + i\omega}{\gamma^2 + 2i\gamma\omega - \omega^2 + \omega_0^2}.$$

Now note the identity

$$\gamma + i\omega = i(\omega - i\gamma), \qquad \gamma^2 + 2i\gamma\omega - \omega^2 + \omega_0^2 = -\Big[(\omega - i\gamma)^2 - \omega_0^2\Big].$$

Therefore

$$X(\omega) = \frac{i(\omega - i\gamma)}{-\big[(\omega - i\gamma)^2 - \omega_0^2\big]} = \frac{1}{i}\,\frac{\omega - i\gamma}{(\omega - i\gamma)^2 - \omega_0^2}.$$

If you include the $(2\pi)^{-1/2}$ normalization in the transform definition you obtain exactly

$$\boxed{X(\omega) = \frac{1}{i\sqrt{2\pi}}\,\frac{\omega - i\gamma}{(\omega - i\gamma)^2 - \omega_0^2}}$$

as stated in the problem (the factor $1/\sqrt{2\pi}$ depends on the chosen FT normalization).

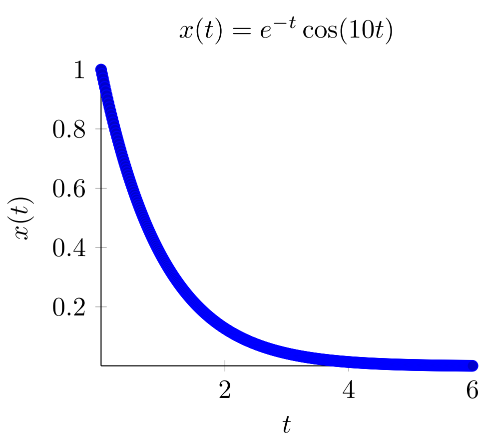
(b) Plots for $\omega_0 = 10$, $\gamma = 1$ and comparison

We plot the time-domain signal $x(t) = e^{-t}\cos(10t)$ for $t \geq 0$ and the frequency-domain power spectral density $|X(\omega)|^2$ using the expression derived above. From the expression computed earlier one convenient closed form for the squared magnitude (no extra normalization factors) is

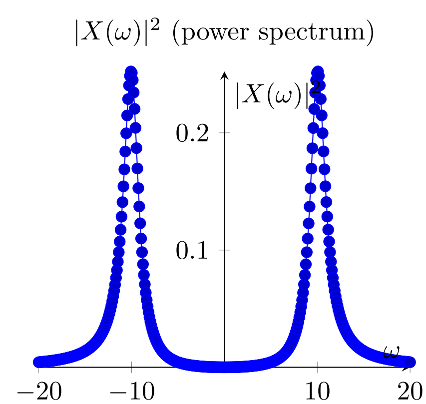
$$|X(\omega)|^2 = \frac{\gamma^2 + \omega^2}{\left(\gamma^2 + \omega^2 - 2\omega\omega_0 + \omega_0^2\right)\left(\gamma^2 + \omega^2 + 2\omega\omega_0 + \omega_0^2\right)}.$$

(You can verify this by multiplying $X(\omega)$ by its complex conjugate.)

Below are two plots produced with `pgfplots` (parameters $\omega_0 = 10$, $\gamma = 1$).



Time domain



Frequency domain

Comparison / interpretation.

- The time-domain plot shows a decaying oscillation at frequency $\approx \omega_0 = 10$ with envelope $e^{-\gamma t} = e^{-t}$. The oscillations die out on a time scale $\sim 1/\gamma$.

- The frequency-domain plot shows two closely related resonance peaks near $\omega = \pm\omega_0$ (because $\cos(\omega_0 t)$ contributes positive- and negative-frequency components). For the chosen parameters the PSD has peaks near $\omega = \pm 10$; the finite damping γ broadens the peaks (a larger γ gives wider, lower peaks). This is the usual time-frequency tradeoff: a rapidly decaying (short) signal produces a broad frequency response; a long-lived oscillator (small γ) produces sharp spectral peaks.

12 Homework 6 - Problem 14

Obtain the Fourier transforms of the following functions.

(a) Hydrogen atom ground and first excited state wavefunctions (position $\to$ momentum)

The hydrogenic wavefunctions (in position space) are

$$\psi_{100}(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}}\, e^{-r/a_0}, \qquad \psi_{200}(\mathbf{r}) = \frac{1}{\sqrt{32\pi a_0^3}}\Big(2 - \frac{r}{a_0}\Big)e^{-r/(2a_0)},$$

where $r = \|\mathbf{r}\|$ and a_0 is the Bohr radius.

We compute the 3D Fourier transform

$$\widetilde{\psi}(\mathbf{k}) = \int_{\mathbb{R}^3} \psi(\mathbf{r})\, e^{-i\mathbf{k}\cdot\mathbf{r}}\, d^3r.$$

Because the position-space wavefunctions are spherically symmetric, the transform depends only on $k = \|\mathbf{k}\|$ and the angular integration can be done by the standard identity

$$\int_{S^2} e^{-i\mathbf{k}\cdot\mathbf{r}}\, d\Omega = 4\pi \frac{\sin(kr)}{kr}.$$

Hence the radial transform reduces to the one-dimensional integral

$$\widetilde{\psi}(k) = 4\pi \int_0^\infty r^2 \psi(r)\, \frac{\sin(kr)}{kr}\, dr = \frac{4\pi}{k}\int_0^\infty r\,\psi(r)\,\sin(kr)\, dr.$$

Ground state, ψ_{100}:

$$\widetilde{\psi}_{100}(k) = \frac{4\pi}{k}\,\frac{1}{\sqrt{\pi a_0^3}}\int_0^\infty r\, e^{-r/a_0}\sin(kr)\, dr.$$

Use the standard definite integral (valid for $a > 0$ and real k)

$$\int_0^\infty r\, e^{-r/a}\sin(kr)\, dr = \frac{2a^3 k}{\big(1 + a^2k^2\big)^2}.$$

With $a = a_0$:

$$\widetilde{\psi}_{100}(k) = \frac{4\pi}{k}\,\frac{1}{\sqrt{\pi a_0^3}} \cdot \frac{2a_0^3 k}{\big(1 + a_0^2k^2\big)^2} = \frac{8\sqrt{\pi}\, a_0^{3/2}}{\big(1 + a_0^2k^2\big)^2}.$$

So (without extra Fourier-normalization factors)

$$\boxed{\widetilde{\psi}_{100}(k) = \frac{8\sqrt{\pi}\, a_0^{3/2}}{\big(1 + a_0^2k^2\big)^2}}$$

(If you use the momentum variable $\mathbf{p} = \hbar\mathbf{k}$ and a different normalization of the Fourier transform, include the corresponding Jacobian/factors.)

First excited state (2s), ψ_{200}:

$$\widetilde{\psi}_{200}(k) = \frac{4\pi}{k}\,\frac{1}{\sqrt{32\pi a_0^3}}\int_0^\infty r\Big(2 - \frac{r}{a_0}\Big)e^{-r/(2a_0)}\sin(kr)\,dr.$$

Evaluate the radial integral (again using standard integrals). With $a \equiv a_0$ one finds

$$\int_0^\infty r\Big(2 - \frac{r}{a}\Big)e^{-r/(2a)}\sin(kr)\,dr = \frac{64a^3k\,(4a^2k^2 - 1)}{(4a^2k^2 + 1)^3}\cdot\frac{1}{k} = \frac{64a^3(4a^2k^2 - 1)}{(4a^2k^2 + 1)^3}.$$

(That simplification comes from evaluating the integral term-by-term and combining common factors.) Hence

$$\widetilde{\psi}_{200}(k) = \frac{4\pi}{k}\frac{1}{\sqrt{32\pi a^3}}\cdot\frac{64a^3(4a^2k^2 - 1)}{(4a^2k^2 + 1)^3} = 64\sqrt{\frac{\pi}{2}}\,a^{3/2}\,\frac{4a^2k^2 - 1}{\big(4a^2k^2 + 1\big)^3},$$

so

$$\boxed{\widetilde{\psi}_{200}(k) = 64\sqrt{\frac{\pi}{2}}\,a_0^{3/2}\,\frac{4a_0^2k^2 - 1}{\big(4a_0^2k^2 + 1\big)^3}}$$

again up to the overall Fourier normalization you adopt.

Remarks:

- Both momentum-space wavefunctions are real (for the chosen FT convention) and spherically symmetric (depend only on k).
- These closed-form expressions are useful to compute momentum distributions $|\widetilde{\psi}(k)|^2$ and expectation values of momentum-dependent operators.

(b) 1D harmonic oscillator, ground and first excited states

The 1D harmonic oscillator position-space eigenfunctions (using oscillator length $a_h = \sqrt{\hbar/(m\omega)}$) are

$$\phi_0(x) = \frac{1}{\pi^{1/4}\,a_h^{1/2}}\,e^{-x^2/(2a_h^2)}, \qquad \phi_1(x) = \sqrt{\frac{2}{\sqrt{\pi}\,a_h^3}}\,x\,e^{-x^2/(2a_h^2)}.$$

We compute the one-dimensional Fourier transform (again using the convention $F(k) = \int_{-\infty}^{\infty} f(x)e^{-ikx}\,dx$).

Ground state ϕ_0: The Gaussian transform is standard:

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}\,e^{-ikx}\,dx = \sigma\sqrt{2\pi}\,e^{-\frac{\sigma^2k^2}{2}}.$$

Here $\sigma = a_h$. Therefore

$$\Phi_0(k) = \frac{1}{\pi^{1/4}a_h^{1/2}}\cdot a_h\sqrt{2\pi}\,e^{-\frac{a_h^2k^2}{2}} = \big(2\pi\big)^{1/2}\,\frac{a_h^{1/2}}{\pi^{1/4}}\,e^{-\frac{a_h^2k^2}{2}}.$$

If you absorb $(2\pi)^{1/2}$ into your FT normalization you will get the more common unit-normalized momentum-space harmonic-oscillator eigenfunction

$$\widetilde{\phi}_0(k) = \Big(\frac{a_h}{\pi^{1/2}}\Big)^{1/2}e^{-a_h^2k^2/2},$$

or, with the symmetric $(2\pi)^{-1/2}$ FT normalization,

$$\Phi_0(k) = \frac{1}{\pi^{1/4} a_h^{1/2}}\, e^{-a_h^2 k^2/2}.$$

(Keep track of which FT convention your course uses; the functional dependence is always Gaussian.)

First excited state ϕ_1: Use the property that multiplication by x in position space corresponds to i times the derivative w.r.t. k in Fourier space:

$$\mathcal{F}\{xf(x)\}(k) = i\,\frac{d}{dk}\mathcal{F}\{f\}(k).$$

Since $\phi_1(x) = \sqrt{\frac{2}{\sqrt{\pi}\,a_h^3}}\,x\,e^{-x^2/(2a_h^2)}$, write it as

$$\phi_1(x) = C\,x\,e^{-x^2/(2a_h^2)} \quad \text{with} \quad C = \sqrt{\frac{2}{\sqrt{\pi}\,a_h^3}}.$$

Thus

$$\Phi_1(k) = C\cdot i\frac{d}{dk}\Big(\text{FT of } e^{-x^2/(2a_h^2)}\Big) = C\cdot i\frac{d}{dk}\Big(a_h\sqrt{2\pi}e^{-a_h^2k^2/2}\Big).$$

Differentiating gives a factor $(-a_h^3 k)\sqrt{2\pi}e^{-a_h^2k^2/2}$ (modulo signs); collecting factors yields (up to FT-normalization conventions)

$$\boxed{\Phi_1(k) \propto i\,k\,e^{-a_h^2k^2/2}}$$

more precisely (with the same normalization choices as for Φ_0)

$$\Phi_1(k) = i\sqrt{\frac{2}{\sqrt{\pi}\,a_h^3}}\;a_h\sqrt{2\pi}\;(-a_h^2k)\,e^{-a_h^2k^2/2} = i\cdot(\text{constant})\;k\,e^{-a_h^2k^2/2},$$

i.e. the momentum-space first excited state is an odd function proportional to $ike^{-a_h^2k^2/2}$. The factor i indicates that the position-space odd real function transforms to an imaginary (odd) function in k-space with the same parity properties.

Summary of key boxed results

- For $x(t) = e^{-\gamma t}\cos(\omega_0 t)\theta(t)$ with FT $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-i\omega t}dt$,

$$X(\omega) = \frac{1}{2}\left[\frac{1}{\gamma + i(\omega-\omega_0)} + \frac{1}{\gamma + i(\omega+\omega_0)}\right] = \frac{1}{i}\,\frac{\omega - i\gamma}{(\omega-i\gamma)^2 - \omega_0^2}.$$

- Hydrogen 1s (no overall normalization constants from FT):

$$\widetilde{\psi}_{100}(k) = \frac{8\sqrt{\pi}\,a_0^{3/2}}{\left(1+a_0^2k^2\right)^2}.$$

- Hydrogen 2s:

$$\widetilde{\psi}_{200}(k) = 64\sqrt{\frac{\pi}{2}}\,a_0^{3/2}\;\frac{4a_0^2k^2-1}{\left(4a_0^2k^2+1\right)^3}.$$

- Harmonic oscillator:
$$\Phi_0(k) \propto e^{-a_\hbar^2 k^2/2}, \qquad \Phi_1(k) \propto i\,k\,e^{-a_\hbar^2 k^2/2}.$$

Notes on conventions: If your class uses the unitary Fourier transform with factors $(2\pi)^{-1/2}$ or uses $\mathbf{p} = \hbar\mathbf{k}$ as the momentum variable, apply those factors and the Jacobian $\mathbf{k} = \mathbf{p}/\hbar$ to convert the expressions to the desired normalization/variable.

If you would like, I can:

1. produce the same derivations but with the $(2\pi)^{-3/2}$ (quantum standard) normalization for the 3D transforms so that momentum-space wavefunctions are normalized in p-space; or
2. provide high-resolution numerical plots of the hydrogen momentum distributions $|\widetilde{\psi}(k)|^2$ vs. k (linear or log scale).
